

Accelerated Root-Finding via Quadrature-Based Newton Variants

A Comparative Study, Numerical Simulation & Methodological Extension

Base Paper Reference

"A Variant of Newton's Method with Accelerated Third-Order Convergence"
S. Weerakoon & T. G. I. Fernando — **Applied Mathematics Letters**, Vol. 13, No. 8 (2000)

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Base Paper Overview: Geometric Core (Standard NM)

Standard Newton's Method (NM)

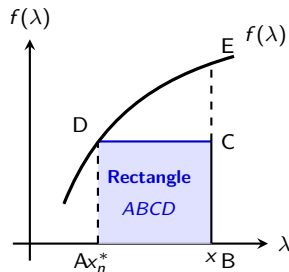
Quadratic Convergence (Order 2)

Approximates $\int_{x_n^*}^x f'(\lambda) d\lambda$ using the **Rectangle Rule** (Left-endpoint approximation):

$$\int_{x_n^*}^x f'(\lambda) d\lambda \approx f'(x_n^*)(x - x_n^*)$$

Area Interpretation

The integral is approximated by the area of the rectangle $ABCD$, ignoring the curve's curvature completely.



Base Paper Overview: Geometric Core (Proposed VNM)

Proposed Variant (VNM)

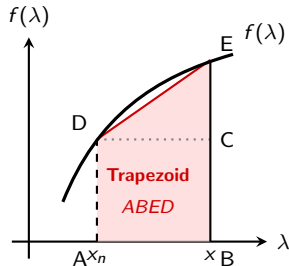
Cubic Convergence (Order 3)

Approximates $\int_{x_n}^x f'(\lambda) d\lambda$ using the **Trapezoidal Rule**:

$$\int_{x_n}^x f'(\lambda) d\lambda \approx \frac{1}{2}(x - x_n) [f'(x_n) + f'(x)]$$

Area Interpretation

The integral is approximated by the trapezoid $ABED$, capturing local slope change (f'') across the interval.



Key Algorithmic Breakthrough

No Second Derivatives (f'') Needed: Solves implicitness via a **Predictor-Corrector** step (x_{n+1}^*), capturing curvature while maintaining low evaluation cost!

Problem Statement & Scope: Phase 1 & 2

Core Problem Statement

Standard 2nd-order Newton's Method (NM) suffers from slow convergence or breakdown near flat slopes ($f'(x) \approx 0$). Higher-order schemes usually require expensive second derivatives (f''). We need an accelerated scheme that balances speed without derivative overhead.

Phase 1 & 2: Base Implementation & Paper Replication

Core Objectives:

- Implement standard NM and proposed VNM algorithms in Python.
- Build iterative loops with strict stopping thresholds ($\varepsilon = 10^{-15}$).

Key Deliverables:

- Replicate Paper's benchmark results across all kind of test functions.
- Track total iterations (i) and Number of Function Evaluations (NOFE).

Scope Progress

Current Focus: Establishing a rock-solid, verified baseline against published literature.

Phase 3: Exploratory Study & Deep Benchmarking

Empirical Order Verification:

- Compute empirical Computational Order of Convergence (COC):

$$\rho \approx \frac{\ln |(x_{n+1} - \alpha)/(x_n - \alpha)|}{\ln |(x_n - \alpha)/(x_{n-1} - \alpha)|}$$

- Prove numerical order $\rho \approx 3.0$ for VNM.

Stress-Testing & Visualization:

- Test algorithm stability under poor initial guesses ($x_0 \in [-10, 10]$).
- Plot semilog error decay curves: $\log_{10} |x_n - \alpha|$ vs. iteration count (n).

Comparative Algorithms

Benchmark VNM against other classic root-finders: **Standard Newton**, **Secant Method**, and **Halley's Method**.

Problem Statement & Scope: Phase 4

Phase 4: Methodological Extension Track

Proposed Extension Idea:

- Replace Trapezoidal Rule with **Simpson's 1/3 Rule**.
- Multi-stage predictor to estimate both midpoint and endpoint.

Target Outcomes:

- Investigate if convergence jumps to **4th or 5th order**.
- Analyze trade-off: extra function evaluations vs. speed gain.

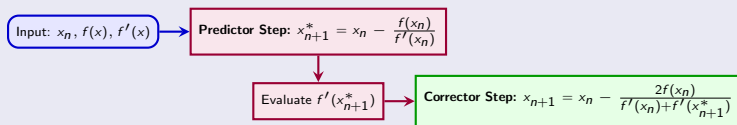
Simpson's 1/3 Rule Quadrature Formula

$$\int_{x_n}^x f'(\lambda) d\lambda \approx \frac{x - x_n}{6} \left[f'(x_n) + 4f' \left(\frac{x_n + x}{2} \right) + f'(x) \right]$$

Final Project Verdict

Chosen Scope: *Exploratory Benchmarking (Phase 3) + Methodological Extension via Simpson's Quadrature (Phase 4).*

Numerical Implementation Pipeline (VNM Solver)



Simulation Control & Tolerance

The numerical loop terminates when both stopping criteria are satisfied ($\epsilon = 10^{-15}$):

1. $|x_{n+1} - x_n| < \sqrt{\epsilon}$
2. $|f(x_{n+1})| < \sqrt{\epsilon}$

Ensures high precision while preventing division-by-zero.

Methodology Benchmarking Metrics

Comparative performance across solvers (NM vs. VNM vs. Simpson's Variant):

- **Iteration Count (i):** Convergence speed to root α .
- **NOFE:** Total evaluations of $f(x)$ and $f'(x)$.
- **Empirical Order (ρ):** Verified numerically via:

$$\rho \approx \frac{\ln |(x_{n+1} - \alpha)/(x_n - \alpha)|}{\ln |(x_n - \alpha)/(x_{n-1} - \alpha)|}$$

Thank You!